Homework 3: Problem 3.7 (Muscle Mass)

Applied Linear Statistical Models, 5th ed. (Kutner, Nachtsheim, Neter & Li), Chapter 3: Diagnostics and Remedial Measures

Name: _____ Course: _____ Date: _____

Setting and fitted model

Data (Problem 1.27). A nutritionist randomly selected 15 women from each 10-year age group, beginning with age 40 and ending with age 79 (n = 60). X is age in years and Y is a measure of muscle mass. Problem 3.7 checks the simple linear regression model (2.1), $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$, with ε_i independent $N(0, \sigma^2)$. The analysis follows the steps of the SAS program in Appendix B. The question text in the shaded boxes is paraphrased from the problem.

Fitted regression function: $\hat{Y} = 156.347 - 1.190X$. Muscle mass is estimated to fall by about 1.19 units for each additional year of age.

Parameter	Estimate	Std. Error	t value	Pr > |t|
Intercept (β_0)	156.3466	5.5123	28.36	<.0001
Age X (β_1)	−1.1900	0.0902	−13.19	<.0001

Source	DF	Sum of Squares	Mean Square	F value	Pr > F
Model	1	11,627.49	11,627.49	174.06	<.0001
Error	58	3,874.45	66.80		
Corrected Total	59	15,501.93			

Table 1. Regression of Y on X. Root MSE = 8.1732, R^2 = 0.7501, Adj. R^2 = 0.7458.

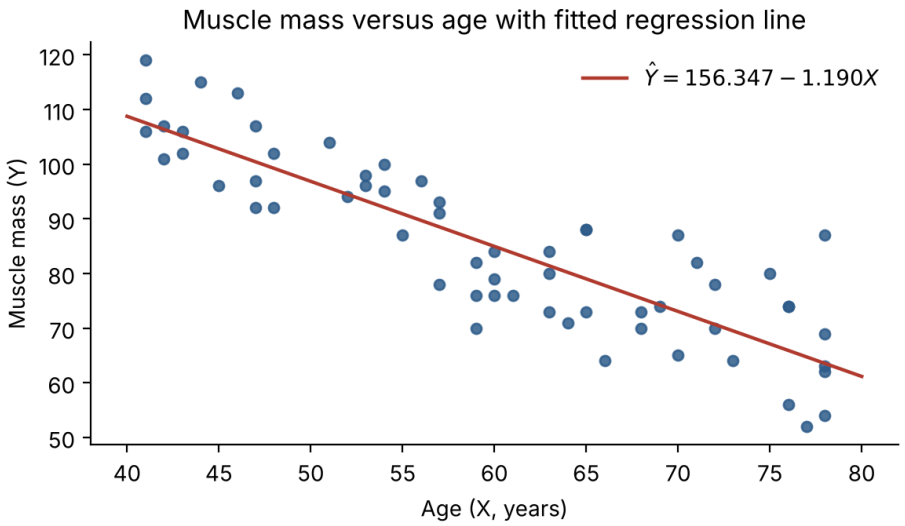


Figure 1. Scatter plot of muscle mass against age with the fitted least squares line.

(a) Stem-and-leaf plot of the ages

Prepare a stem-and-leaf plot for the ages X_i. Is this plot consistent with the random selection of women from each 10-year age group? Explain.

```
Stem | Leaf                 #          Stem | Leaf                 #
-----+----------------------------     -----+--------------------------
  7  | 0012235666788888     16           7  | 5666788888          10
  6  | 000133345556889      15           7  | 001223               6
  5  | 12334456777999       14           6  | 5556889              7
  4  | 111223345677788      15           6  | 00013334             8
-----+----------------------------       5  | 56777999             8
  Key: 4 | 1 = 41 years                   5  | 123344               6
                                          4  | 5677788              7
                                          4  | 11122334             8
                                         -----+--------------------------
                                           Key: 4 | 1 = 41 years
```

Figure 2. Stem-and-leaf plot of age. Left: one stem per decade. Right: split stems (upper line = leaves 5–9, lower line = leaves 0–4).

Age group	40–49	50–59	60–69	70–79	Total
Number of women	15	14	15	16	60
Design (Problem 1.27)	15	15	15	15	60

Table 2. Frequencies by 10-year age group, read off the stems.

Interpretation. The four stems (40s, 50s, 60s, 70s) have almost the same number of leaves (15, 14, 15, 16), so the plot looks like a **roughly uniform** distribution of ages from 41 to 78. That is what we expect when about the same number of women is drawn from each 10-year group. Within each stem the leaves are spread across most of the decade (for example 41–48, 51–59, 60–69 and 70–78), with no heavy clustering at a single age and no large gaps, which is consistent with women being selected *at random* within each group. The one-woman difference between the 50s (14) and the 70s (16) is minor and does not change the overall picture.

Conclusion: Yes. The stem-and-leaf plot is consistent with random selection of (about) 15 women from each 10-year age group from 40 to 79.

(b) Residuals and dot plot

Obtain the residuals e_i and prepare a dot plot of the residuals. What does this plot show?

The residuals $e_i = Y_i - \hat{Y}_i$ were saved with the OUTPUT statement of PROC REG. The full list of all 60 residuals is in Appendix A. Their summary is:

n	Mean	Median	Std. Dev.	Minimum	Maximum	Skewness
60	0.000	−0.597	8.104	−16.137	23.473	0.328

Table 3. Summary of the residuals.



Figure 3. Dot plot of the 60 residuals. Each dot is one residual rounded to the nearest unit.

What the plot shows.

- The residuals are **centered at zero** (mean exactly 0, median −0.60) and spread fairly **symmetrically** on both sides of 0. Most fall between about −16 and +14.
- The shape is roughly **mound-shaped**, with no gaps or clusters. This agrees with normally distributed errors with a common variance.
- One residual stands apart: observation 53 (X = 78, Y = 87, $\hat{Y}$ = 63.53) has e = 23.47. Its semistudentized residual is $e^* = e/\sqrt{\text{MSE}} = 23.47/8.173 = 2.87$. It is the largest residual and a possible mild outlier, but it is well below the common rule of thumb $|e^*| \geq 4$ for flagging outliers.

Conclusion: The dot plot shows no serious departure from model assumptions. The residuals are centered at 0, roughly symmetric and bell-shaped, with one somewhat large positive residual (observation 53) that should be watched.

(c) Residual plots against $\hat{Y}$ and against X

Plot the residuals e_i against $\hat{Y}_i$ and also against X_i on separate graphs to ascertain whether any departures from regression model (2.1) are evident. State your findings.

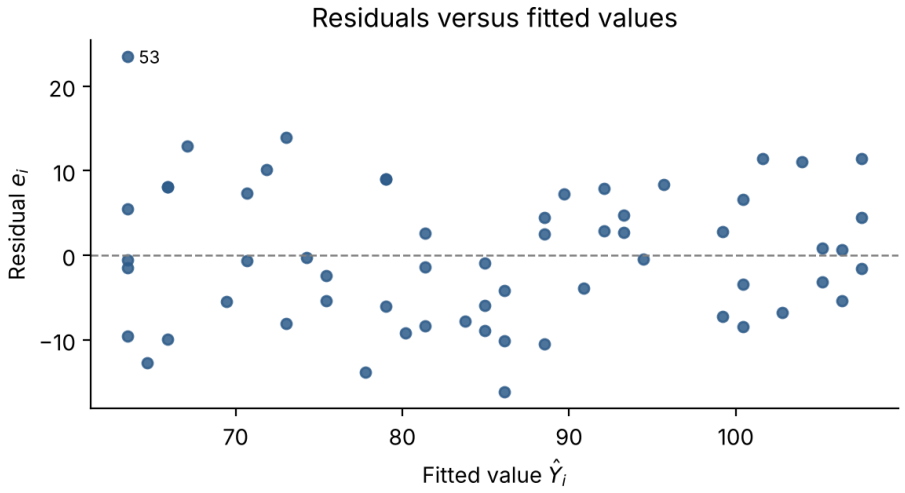


Figure 4. Residuals versus fitted values $\hat{Y}_i$ (dashed line at e = 0).

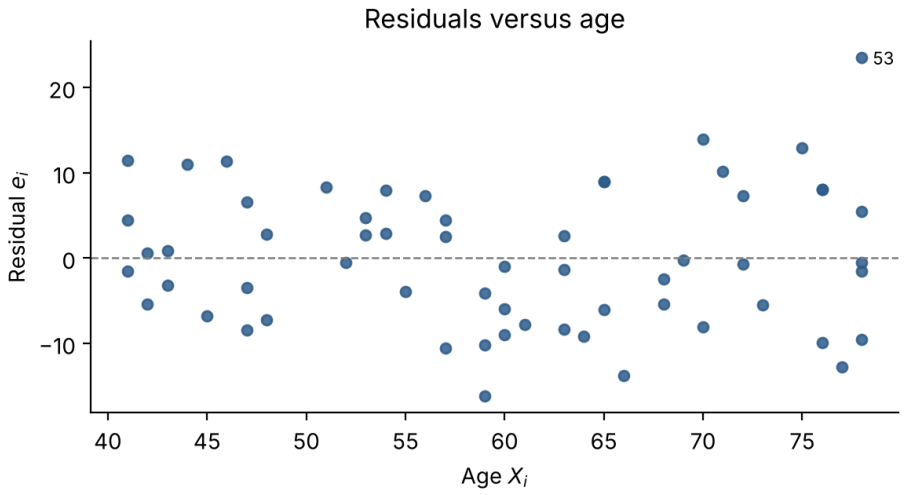


Figure 5. Residuals versus age X_i (dashed line at e = 0).

Age group	40–49	50–59	60–69	70–79
Mean residual	0.88	−0.32	−3.32	2.57
Std. dev. of residuals	6.89	7.62	6.54	10.24

Table 4. Residual mean and spread within each age group (supporting the visual reading of Figure 5).

Findings. Because $b_1 < 0$, $\hat{Y}$ is a decreasing linear function of X. Figure 4 is therefore a mirror image of Figure 5, and the two plots carry the same information. We check each type of departure from model (2.1) that these residual plots can reveal:

- **Nonlinearity of the regression function:** none. The residuals form a horizontal band around 0 with no curved (U-shaped or inverted-U) pattern. The mean residual in each age group is close to 0 (Table 4) and changes sign with no systematic trend. A linear regression function is appropriate.
- **Nonconstancy of the error variance:** no clear funnel or megaphone shape. The band has about the same width over the whole range. The spread is a little larger for the oldest women (70–79, small $\hat{Y}$), mainly because of observation 53. This is mild and is tested formally in part (e).
- **Outliers:** no extreme outliers. Observation 53 ($e = 23.47$, $e^* \approx 2.87$) is the most distant point but is not beyond $\pm 4\sqrt{MSE}$.
- **Nonindependence / omitted variables:** the data are not in time order, so these plots cannot show serial correlation. They show no structure that would point to it.

Conclusion: No important departures from regression model (2.1) are evident. The linear form is adequate and the error variance looks roughly constant, with at most a slight increase in spread at the oldest ages driven largely by one observation.

(d) Normal probability plot and test for normality

Prepare a normal probability plot of the residuals and test the reasonableness of the normality assumption. (As instructed, the normal probability plot and the Shapiro–Wilk test from PROC UNIVARIATE are used.)

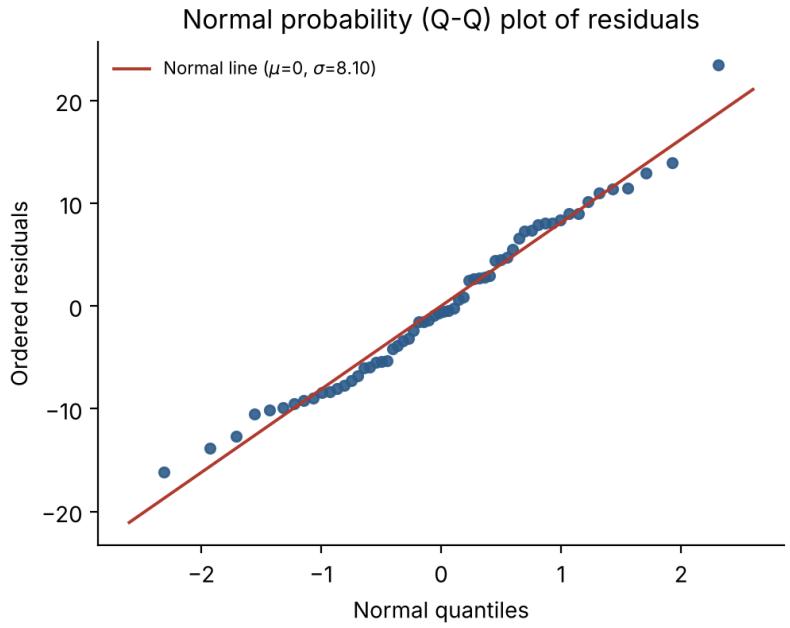


Figure 6. Normal probability plot of the residuals; the reference line uses the estimated mean and standard deviation (mu=est, sigma=est).

Reading the plot. The ordered residuals fall close to a straight line through the middle of the distribution. The ends bend a little: the largest residual (observation 53) sits above the line and the smallest residuals sit slightly above it as well. This suggests a slight right skew (sample skewness 0.33), but there is no S-shape and the departure from linearity is small. The plot looks like it came from a normal distribution.

Hypotheses (α = 0.05):

H_0: the error terms ε_i are normally distributed.
H_a: the error terms ε_i are not normally distributed.

Test for normality	Statistic	p value
Shapiro–Wilk	W = 0.97958	Pr < W = 0.4112

Table 5. Shapiro–Wilk test applied to the residuals (the test requested by the NORMAL option of PROC UNIVARIATE).

Decision rule: reject H_0 if Pr < W is less than α = 0.05. Here Pr < W = 0.4112 > 0.05 (it is also far above α = 0.01 and α = 0.10), so we **fail to reject H_0**.

Supporting check (correlation test). The coefficient of correlation between the ordered residuals and their expected values under normality, $\sqrt{MSE} \cdot z[(k - .375)/(n + .25)]$, is $r = 0.9897$. Large values of r support normality. A Monte Carlo simulation (200,000 normal samples of size 60, same plotting positions) gives critical values of about 0.971 for α = 0.01 and 0.980 for α = 0.05; these should be compared with Table B.6 in the text. Since $r = 0.990$ exceeds them, this check agrees with the Shapiro–Wilk test.

Conclusion: We cannot reject H_0 (W = 0.9796, p = 0.4112). The normal probability plot is nearly linear, and the assumption that the error terms are normally distributed is reasonable for these data.

(e) Breusch–Pagan test for constancy of error variance

Conduct the Breusch–Pagan test to determine whether or not the error variance varies with the level of X. Use α = .01. State the alternatives, decision rule, and conclusion. Is your conclusion consistent with your preliminary findings in part (c)?

The Breusch–Pagan test assumes the error variance is related to X through $\log_e \sigma^2_i = \gamma_0 + \gamma_1 X_i$. Following the SAS program, the squared residuals e^2_i are regressed on X_i (auxiliary regression).

Alternatives (α = 0.01):

H_0: $\gamma_1 = 0$ (the error variance is constant; it does not depend on X).
H_a: $\gamma_1 \neq 0$ (the error variance changes with the level of X).

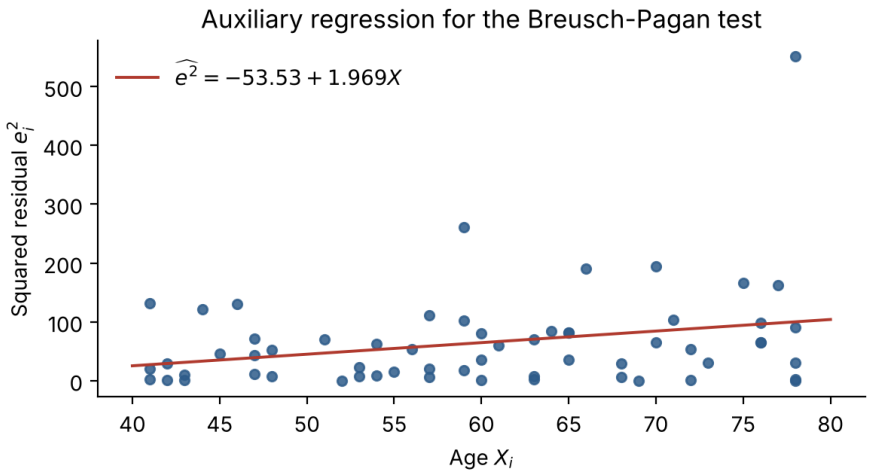


Figure 7. Squared residuals against age with the fitted auxiliary regression line.

Auxiliary regression of e² on X	Value
Fitted equation	ê² = −53.533 + 1.9690 X
SSR* (regression sum of squares)	31,833.43
SSE* (error sum of squares)	400,088.77
R²* of the auxiliary regression	0.07370
n	60
SSE from original regression	3,874.447

Table 6. Auxiliary regression of the squared residuals on X (model e2 = X in the SAS program).

Test statistic computed by the SAS program (Step 5, BP = nR²*):

BP = nR²* = 60 × 0.07370 = **4.4221**, df = 1, p value = P(χ^2_1 > 4.422) = **0.0355**.

n	Rsq	BP	df	p_value
60	0.07370	4.42211	1	0.03548

Table 7. Breusch–Pagan test statistic and p value (the quantities computed in Steps 4–6 of the SAS program).

Textbook form of the statistic, for reference: X^2_{BP} = (SSR*/2) ÷ (SSE/n)² = (31,833.43/2) ÷ (3,874.447/60)² = 15,916.71 ÷ 4,169.82 = **3.817** (p = 0.0507). Both versions lead to the same decision.

Decision rule: with α = 0.01, the critical value is $\chi^2(0.99; 1) = 6.635$. If BP ≤ 6.635 (equivalently p ≥ 0.01), conclude H_0. If BP > 6.635 (p < 0.01), conclude H_a.

Decision: BP = 4.422 ≤ 6.635 and p = 0.0355 > 0.01 (textbook form: 3.817 ≤ 6.635, p = 0.0507). We **fail to reject H_0**. (Note: at α = 0.05 the nR^{2*} version, p = 0.0355, would reject H_0, while the textbook form, p = 0.0507, would not; the decision here uses α = 0.01 as the problem specifies.)

Conclusion: At α = 0.01 there is not enough evidence that the error variance varies with age. We conclude the error variance is constant ($\gamma_1 = 0$).

Consistency with part (c). Yes, the result is consistent. The residual plots in part (c) showed no funnel shape and a band of roughly constant width, with only a slightly wider spread for the oldest women. The positive auxiliary slope (1.97) and moderately small p value reflect that slight widening, but it is not significant at α = 0.01. The widening is driven mostly by observation 53: without it, nR^{2*} drops to 2.43 (p = 0.119).

Summary

Part	Result	Conclusion
(a)	Leaves per decade: 15, 14, 15, 16	Consistent with random selection from each age group
(b)	Residuals centered at 0, symmetric; max e = 23.47 (obs. 53)	No serious departures; one mild outlier
(c)	Random horizontal band in e vs $\hat{Y}$ and e vs X	Linear model adequate; variance roughly constant
(d)	W = 0.9796, p = 0.4112; r = 0.990	Fail to reject H_0: normality reasonable
(e)	BP = 4.422, p = 0.0355 (X^2BP = 3.817) < $\chi^2(.99;1)$ = 6.635	Fail to reject H_0: constant variance

Appendix A. Fitted values and residuals (output data set muscle_resid)

Obs	Y	X	Ŷ	e	Obs	Y	X	Ŷ	e
1	106	43	105.1768	0.8232	31	73	68	75.4269	−2.4269
2	106	41	107.5567	−1.5567	32	73	63	81.3768	−8.3768
3	97	47	100.4168	−3.4168	33	76	60	84.9468	−8.9468
4	113	46	101.6068	11.3932	34	80	63	81.3768	−1.3768
5	96	45	102.7968	−6.7968	35	84	63	81.3768	2.6232
6	119	41	107.5567	11.4433	36	71	64	80.1869	−9.1869
7	92	47	100.4168	−8.4168	37	64	66	77.8069	−13.8069
8	112	41	107.5567	4.4433	38	88	65	78.9969	9.0031
9	92	48	99.2268	−7.2268	39	79	60	84.9468	−5.9468
10	102	48	99.2268	2.7732	40	88	65	78.9969	9.0031
11	107	42	106.3668	0.6332	41	73	65	78.9969	−5.9969
12	107	47	100.4168	6.5832	42	74	69	74.2369	−0.2369
13	102	43	105.1768	−3.1768	43	76	61	83.7568	−7.7568
14	115	44	103.9868	11.0132	44	87	70	73.0469	13.9531
15	101	42	106.3668	−5.3668	45	70	68	75.4269	−5.4269
16	87	55	90.8968	−3.8968	46	69	78	63.5269	5.4731
17	91	57	88.5168	2.4832	47	54	78	63.5269	−9.5269
18	97	56	89.7068	7.2932	48	62	78	63.5269	−1.5269
19	82	59	86.1368	−4.1368	49	78	72	70.6669	7.3331
20	78	57	88.5168	−10.5168	50	65	70	73.0469	−8.0469
21	95	54	92.0868	2.9132	51	64	73	69.4769	−5.4769
22	98	53	93.2768	4.7232	52	74	76	65.9069	8.0931
23	94	52	94.4668	−0.4668	53	87	78	63.5269	23.4731
24	96	53	93.2768	2.7232	54	63	78	63.5269	−0.5269
25	100	54	92.0868	7.9132	55	82	71	71.8569	10.1431
26	84	60	84.9468	−0.9468	56	80	75	67.0969	12.9031
27	70	59	86.1368	−16.1368	57	52	77	64.7169	−12.7169
28	104	51	95.6568	8.3432	58	56	76	65.9069	−9.9069
29	76	59	86.1368	−10.1368	59	70	72	70.6669	−0.6669
30	93	57	88.5168	4.4832	60	74	76	65.9069	8.0931

Appendix B. SAS program (HW3.sas)

```
data muscle;
    input obs Y X;
    datalines;
    /* 60 data lines (obs Y X) as listed in Appendix A */
;
run;

proc print data=muscle;
run;

/*(a)*/

ods graphics off;

proc univariate data=muscle plot;
    var X;
    title "Stem-and-Leaf Plot of Ages";
run;

ods graphics on;



/* Part (b): Obtain residuals and make a dot plot */

proc reg data=muscle;
    model Y = X;
    output out=muscle_resid
        predicted=Yhat
        residual=e;
run;
quit;

/* Print residuals */
proc print data=muscle_resid;
    var Y X Yhat e;
    title "Residuals from the Regression of Muscle Mass on Age";
run;

/* Dot plot of residuals */
proc sgplot data=muscle_resid;
    dot e;
    xaxis label="Residual";
    title "Dot Plot of Residuals";
run;

/* Part (c): Residual plots */

proc reg data=muscle;
    model Y = X;
    output out=muscle_resid
        predicted=Yhat
        residual=e;
run;
quit;

/* Residuals versus fitted values */
proc sgplot data=muscle_resid;
    scatter x=Yhat y=e;
    refline 0 / axis=y lineattrs=(pattern=shortdash);
    xaxis label="Fitted Values";
    yaxis label="Residuals";
    title "Residuals versus Fitted Values";
run;

/* Residuals versus age */
proc sgplot data=muscle_resid;
    scatter x=X y=e;
    refline 0 / axis=y lineattrs=(pattern=shortdash);
```

```
    xaxis label="Age";
    yaxis label="Residuals";
    title "Residuals versus Age";
run;

/* Part (d): Normal probability plot and Shapiro-Wilk test */

proc reg data=muscle;
    model Y = X;
    output out=muscle_resid
        predicted=Yhat
        residual=e;
run;
quit;

/* Normal probability plot and Shapiro-Wilk test */
proc univariate data=muscle_resid normal;
    var e;
    qqplot e / normal(mu=est sigma=est);
    title "Normal Probability Plot and Shapiro-Wilk Test for Residuals";
run;

/*\(H_0\): residuals are normally distributed.
\(H_a\): residuals are not normally distributed.
If Pr < W is small, for example less than 0.05, reject normality.
If Pr < W is not small, we do not have strong evidence against normality.*/

 /* Part (e): Breusch-Pagan test manually */

/* Step 1: Fit original regression and save residuals */
proc reg data=muscle;
    model Y = X;
    output out=muscle_resid
        predicted=Yhat
        residual=e;
run;
quit;

/* Step 2: Compute squared residuals */
data muscle_bp;
    set muscle_resid;
    e2 = e**2;
run;

/* Step 3: Auxiliary regression: e_i^2 on X */
ods output FitStatistics=fit_bp;

proc reg data=muscle_bp;
    model e2 = X;
    title "Auxiliary Regression for Breusch-Pagan Test";
run;
quit;

ods output close;

/* Step 4: Extract R-square */
data _null_;
    set fit_bp;
    if Label2 = "R-Square" then call symputx("Rsq", nValue2);
run;

/* Step 5: Compute BP = n R^2 and p-value */
proc sql noprint;
    select count(*) into :n
    from muscle_bp
    where e2 is not missing and X is not missing;
quit;

data bp_result;
    n = &n;
    Rsq = &Rsq;
```

```
    BP = n * Rsq;
    df = 1;
    p_value = 1 - probchi(BP, df);
run;

/* Step 6: Print chi-square statistic and p-value */
proc print data=bp_result noobs;
    var n Rsq BP df p_value;
    title "Breusch-Pagan Test Statistic and P-value";
run;

/*\(H_0\): constant error variance.
\(H_a\): error variance depends on \(X\).
If p_value < alpha, reject \(H_0\).
If p_value >= alpha, do not reject \(H_0\).*/
```